

★ Motor controller flow of energy

$$I = V/R, \quad R = \rho \cdot (l/A) \quad I = \frac{V}{\rho} \left(\frac{1}{l} \right) \left(\frac{A}{\rho} \right)$$

resistivity ρ length of $\left(\frac{l}{A} \right) = \left(\frac{1}{\rho} \right) \left(\frac{1}{\rho} \right)$

cross section $J = E \sigma \rightarrow$ conductivity

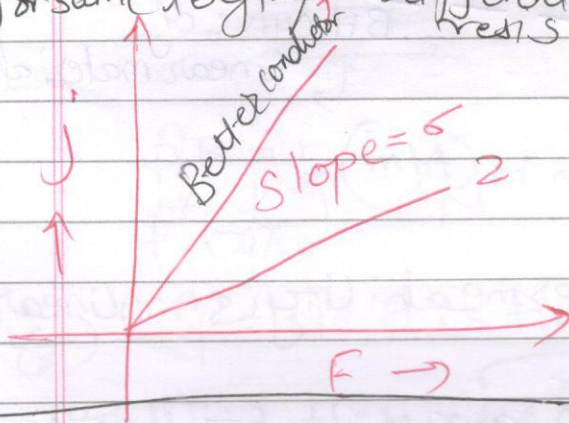
Current density J electric field E

more slope
better conductor.

Compare Cu & Al

for same geometry
different resist.

hence material
for same length $\rightarrow$ different resist.



webster $\Phi = \frac{F}{R}$

mag flux

$B \cdot A$
flux density

mmf

$I \cdot N$

Area

Current

NO. of turns

Reluctance = $l / (\mu A)$

length

permeability = $\mu_0 \cdot \mu_r$

permeability of free space.

$$\Phi \cdot R = F$$

$$\Phi \cdot \left(\frac{l}{\mu A} \right) = I \cdot N$$

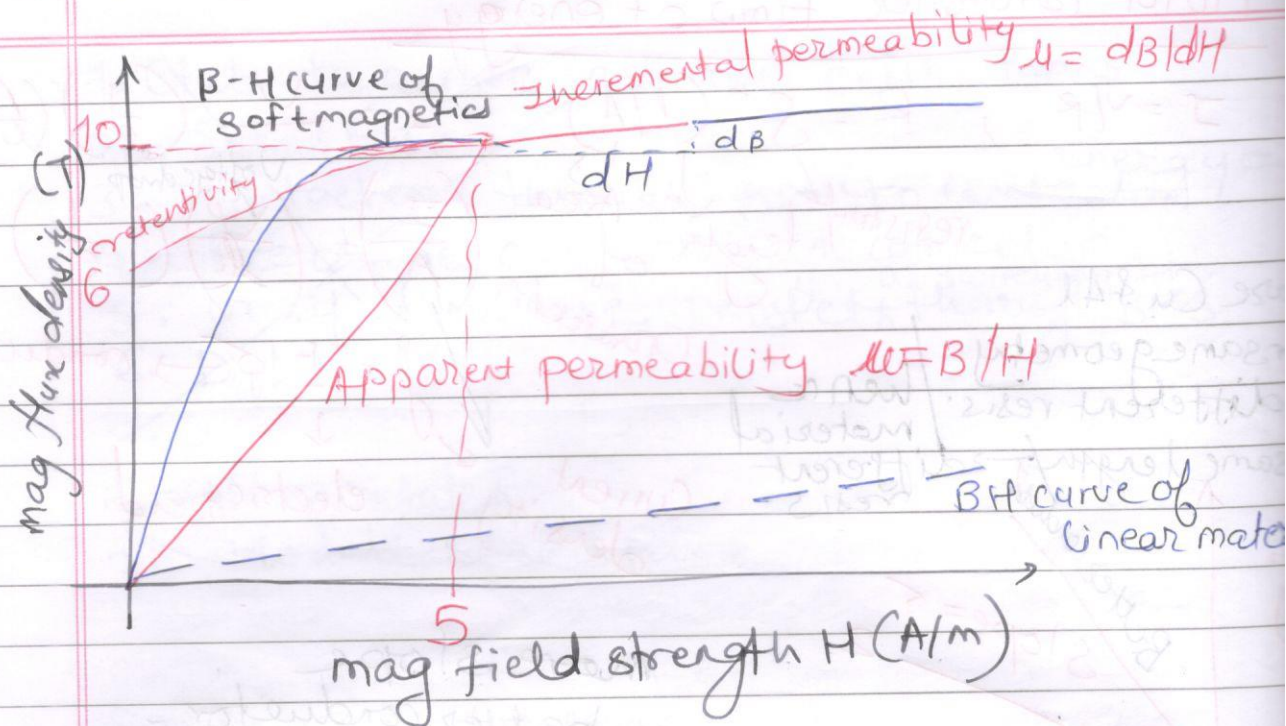
$$\left(\frac{\Phi}{A} \right) = \mu \cdot \left(\frac{I \cdot N}{l} \right)$$

$\frac{wb}{m^2} = I$

Ampere turns / length (meter)

$$B = \mu \cdot H$$

field strength / field strength

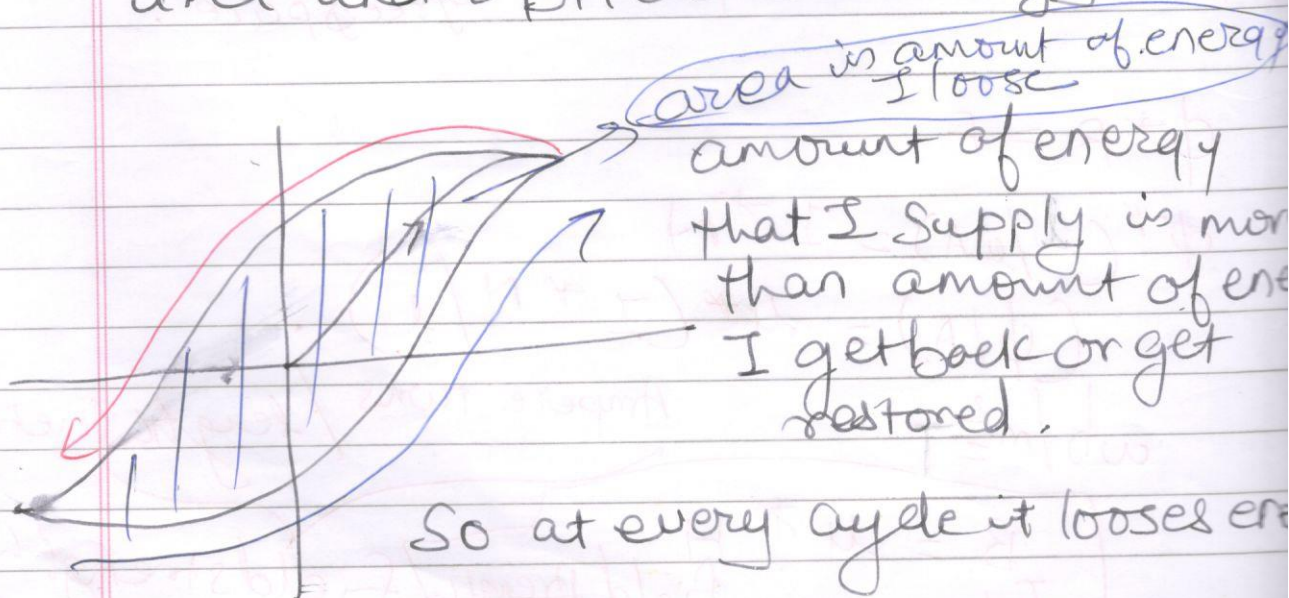


Unlike conductivity, permeability is nonlinear

After the current is taken away it still will have some mag. flux & that is called retentivity.

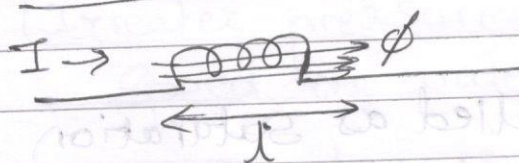
To take out that retentivity I pass -ve current let's say -1, -ve magnetising force is called coercivity.

Product of B & H is power
area under BH curve is energy.



This is Hysteresis loss. Steel gets magnetized & de
 magnetized & we lose energy. (AC
 current with 50 Hz frequency is 50 times
 demagnetizing contributes to more losses)

Flux linkage & inductance.



$$\Phi \cdot R = F$$

$$\Phi \left(\frac{l}{\mu A} \right) = I \cdot N \quad \text{Flux links } N \text{ turns of wire}$$

(3) $\Psi = N \cdot \Phi$ is flux linkage

$$\Psi = N^2 \cdot \left(\frac{\mu A}{l} \right) \cdot I$$

Inductance L

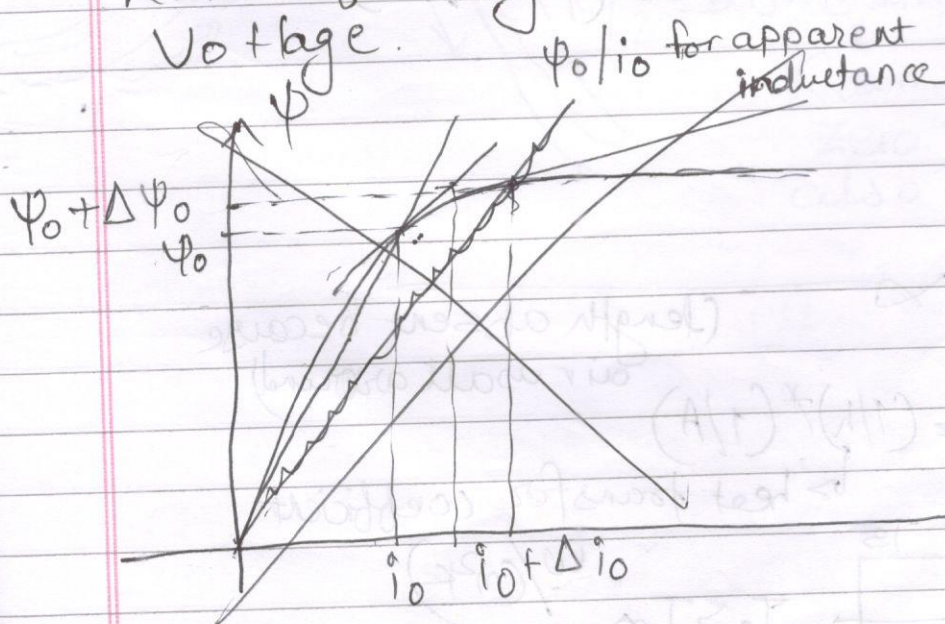
Permeance $P = \left(\frac{\mu A}{l} \right)$

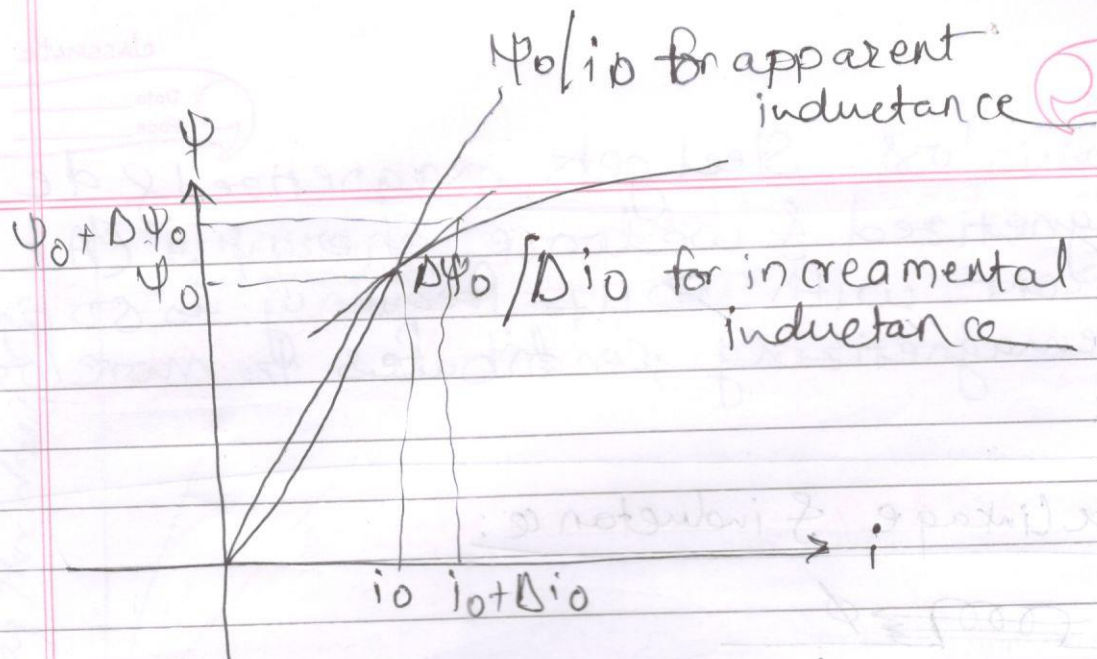
Inductance $L = N^2 \cdot \left(\frac{\mu A}{l} \right)$
 $= N^2 \cdot P$

$$\Psi = L I$$

L & P both are measured in
 Henries 'H'

Rate of change of Ψ will result in formation
 of voltage.

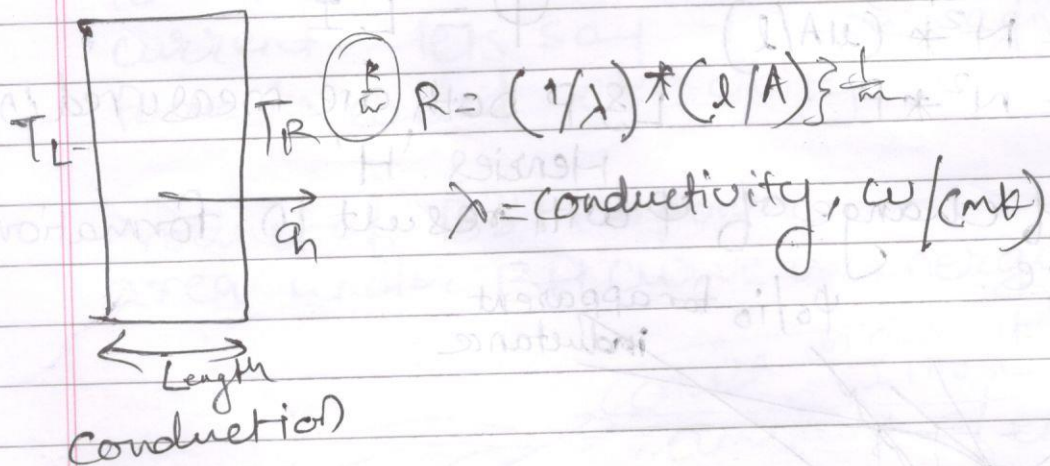




slope tapers $\rightarrow$ called as saturation
beyond certain amount of current there
is no increase in ~~current~~ flux, therefore
flux linkage saturates, so it becomes
poor inductance. affects torque.

Flow of heat

Heat Current $q_h = \Delta T / R$ thermal
heat current in watt $\leftarrow$ $\downarrow$ resist (K/w)
temp. diff. (K)



$T_L > T_R$

convection

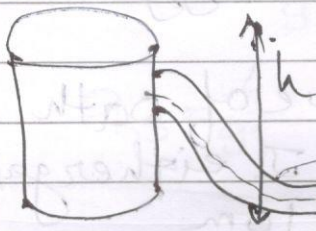
(length absent because
air is all around)

$R = (1/h) * (1/A)$
 $\rightarrow$ heat transfer coefficient
 $W/(m^2K)$

T_∞ T_s q_h

ohm's law

Current

The flow q (in liters/min)

Voltage

is proportional to h (the 'head')

→ Tube offers resis → resis.

If bucket (20 liters) is filled in 5 mins, 4 liters/min.

(Greater pressure on tank which creates speed of water & hence rate)

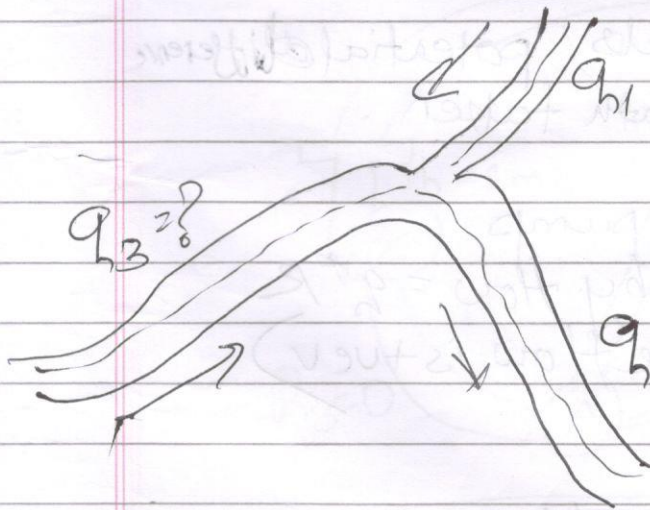
More length → more resis

More area → Less resis.

ohms law $h = R * q$

$$R = \rho * \left(\frac{1}{A}\right)$$

material & geometry.

Kirchoff's first law — mass is conserved. $q_1 = 3$ liters/min $q_3 = -2$ liters/min

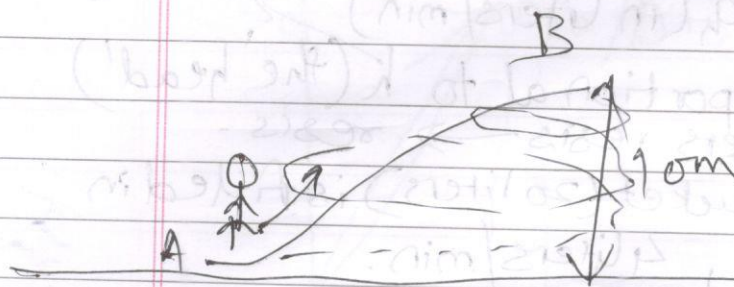
(water is flowing in)

 $q_2 = 5$ liters/min

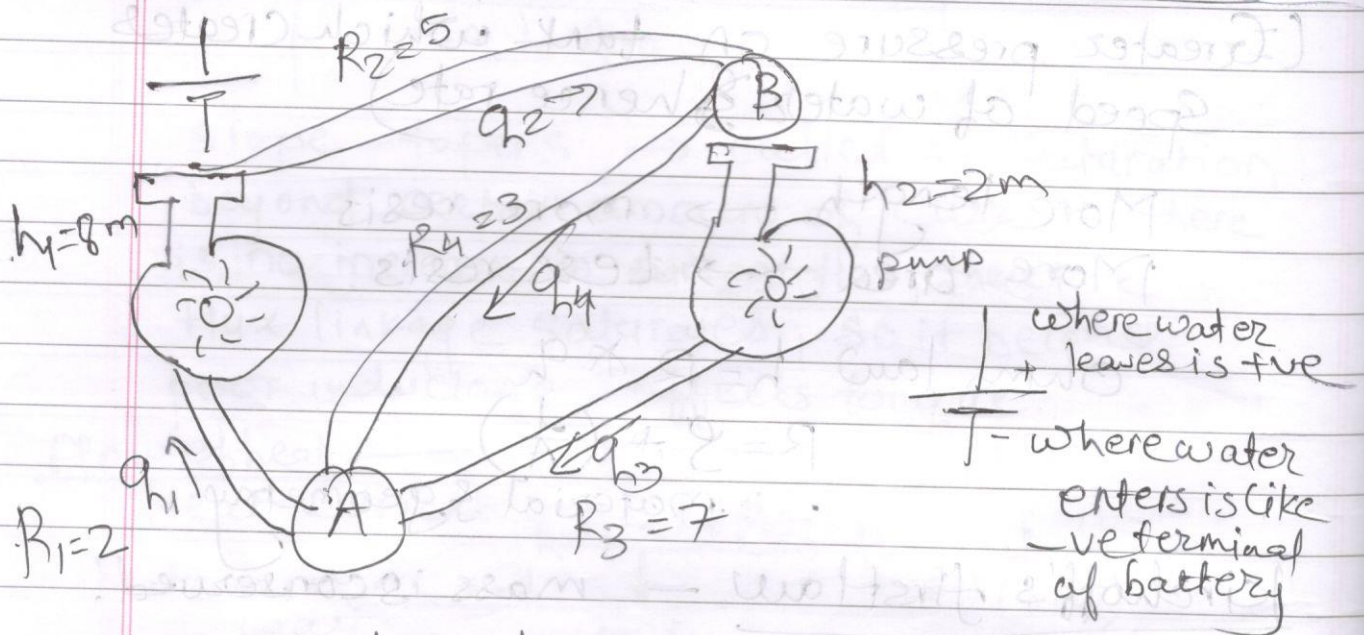
Sum of flow at point is zero

"also called continuity Equation"

Kirchoff's second law :- conservation of energy.



irrespective of path I take, I either gain or lose 10m



Kirchoff's 2nd law tells potential difference does not depend on path taken.

- V is ~~caused~~ caused by pumps
- V can also be caused by flow $= q \cdot R$
(moving against the flow is +ve V)

R in m-min/liter

In case of water flow

$$\frac{\text{head}}{\text{flow rate}} = \frac{\text{meter}}{\text{liters/min}} = \text{meter} \cdot \text{min/liter}$$

against is +ve
towards is -ve

Point A $\rightarrow$ Point B 3 different paths

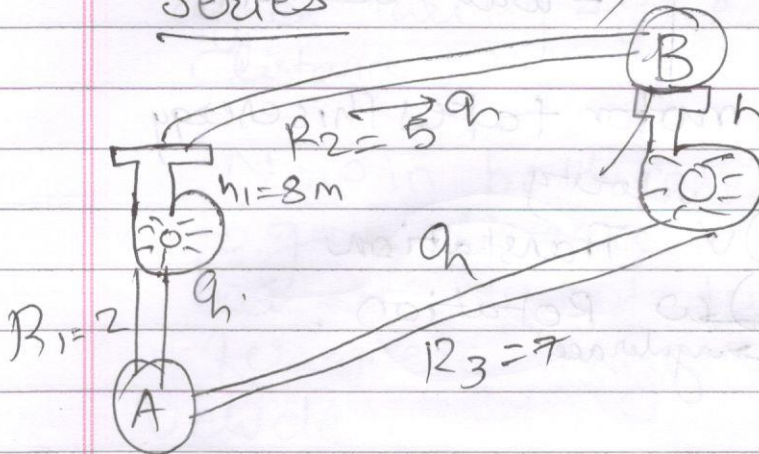
$$V_B - V_A = q_4 R_4 = q_3 R_3 + 2 = -q_1 R_1 + 8 - q_2 R_2$$

first law $\Rightarrow q_1 = q_2$ & $q_1 = q_3 + q_4$

$$q_1 = q_2 = \frac{74}{91}$$

$$q_3 = \frac{4}{91}, \quad q_4 = \frac{10}{13}$$

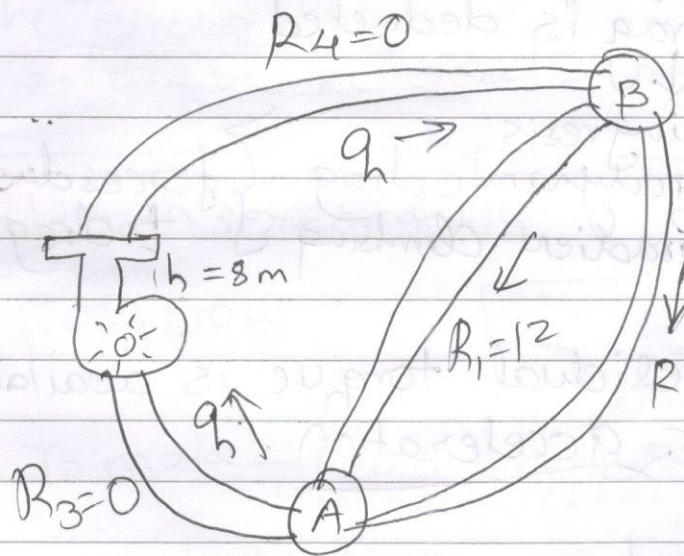
Series



$$R = R_1 + R_2 + R_3$$

$$h_2 = 2m, \quad q = \frac{3}{7}$$

Parallel



$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$q = 2$$

Power & efficiency

force $\times$ flow = power

Electrical

$$P = V \times I$$

$$= I^2 R$$

$$P = I^2 R$$

$$P = V^2 / R$$

$$\text{Energy } E = \int p \cdot dt$$

(Δt is time interval)

E is measured in J (Joules)
= watt-second.

What happen when motor takes this energy

Mechanical

$$P = F \times V = (ma) \overset{\text{velocity}}{V} \text{ Translation}$$

$$P = \tau \times \omega = (I\alpha) \overset{\text{angular acc}^n}{\omega} \text{ Rotation.}$$

moment of inertia

In EV's force F or Torque τ
available for acceleration is net
after all drag is deducted.

Rolling resis
Aerodynamic drag
Gradient climbing } forces due to drag

Whatever residual torque is available
is used for acceleration.

★ Regeneration
26 N-m delivered

→ Dissipative load results in heat
(Heat can not be recovered.)

→ Rolling resis & Air drag

→ Non dissipative : 1) Climbing (pot energy)
2) Acceleration (kin energy)

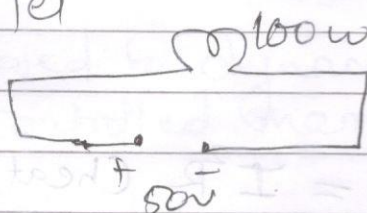
- When I go uphill in motor, potential energy gets stored so using that I can go downhill without starting motor.
- Similarly when I go very fast for some distance (accelerate for some time) then even after turning off the m I can go some distance.

Now in both cases I can use this energy to gain distance without starting m , but it will not to my satisfaction because friction etc might decrease dynamics of vehicle.

So smart way is to use this energy to regenerate the energy.
It's torque reversal - reverse torque drives m .

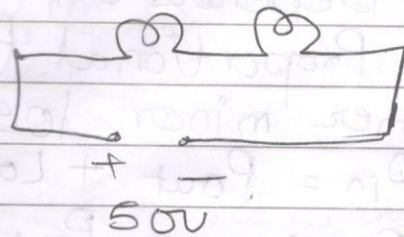
lights in series & ||

50V, 100W



To make room brighter

- 1) how much brighter each bulb will be
- 2) how much brighter room will be



Serial lights: - 1) each bulb has identical Power & voltage.

all bulbs glow same

Even if one blows off, the other will keep on glowing.

How to
wire up
2 bulbs

Electrical to Mechanical Conversion

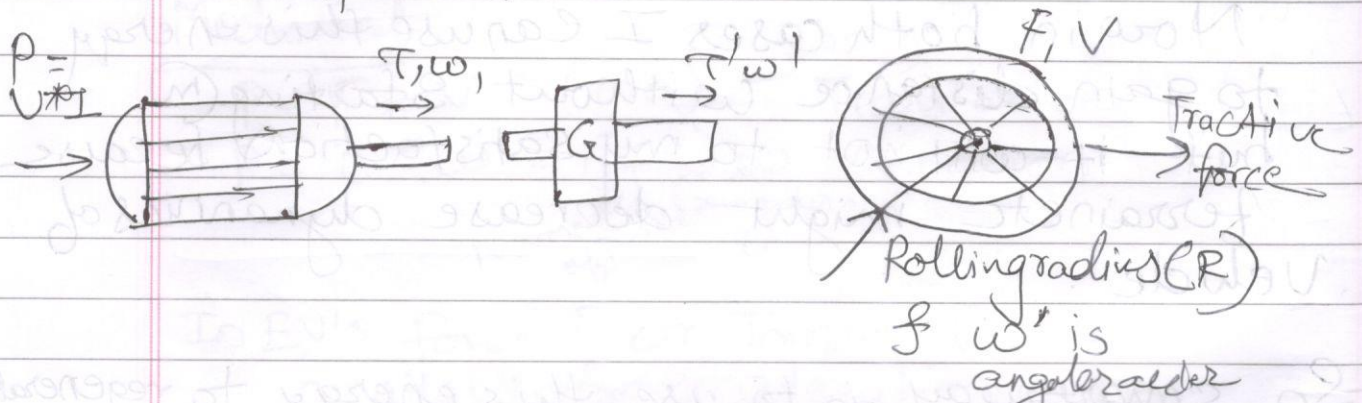
Input :- Electrical Power = $V * I$

Motor Output :- Mechanical Power = $T * \omega$

Gear output :- Mechanical Power

$$= T' * \omega'$$

→ at low ω speed, I want increase in torque o/p.



Vehicle

$$\text{Mch. power} = F * V$$

$$V \text{ Linear Speed} = \frac{\omega' * R}{1} \quad \frac{T'}{R} = F$$

force making it to move

Losses in ω

$$\text{Copper loss} = I^2 R \text{ (heat)}$$

Steel loss

- Depends on material grade
 - Proportional to ω^2 approx.
- Other minor losses } hysteresis
 } eddy current

$$P_{in} = P_{out} + \text{Losses}$$

$$\text{Efficiency} = P_{out} / P_{in}$$

$$\text{Gear ratio} = G \rightarrow \text{output speed is kinematic quantity}$$

$$\omega' = \omega / G$$

Gear efficiency $\approx$ typically 80%.

$$\Rightarrow \tau' \omega' = 0.8 \tau \omega$$

Since it is function of Geometry
SO

$$\text{Traction Force } F = \tau' / r.$$

(M) calculations

A (M) delivers 5 kW at 3500 rpm

- efficiency is 90%.
- half of losses are due to steel, rest due to copper
- coupled to constant load.

+ how much heat is generated?

+ How much power does it consume?

[usually (M) is coupled to dynamometer so test can be done in controlled manner]

- Speed is reduced to 1000 rpm

+ what is efficiency of (M) now?

steel loss goes down as speed

but Cu loss will not change as it depends on I which depends on load. (as load is unchanged here, I is unchanged so Cu loss is const)

- (M) is coupled to gear box of ratio 10, efficiency 80%.

+ why the o/p shaft of gear box is thicker than i/p

shaft is thinner on i/p side on gear box

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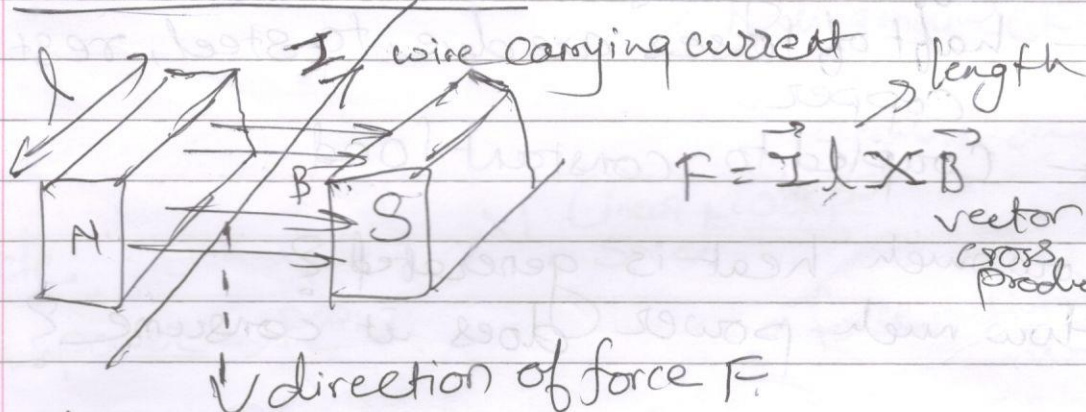
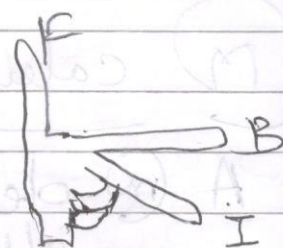
usually we reduce the speed using gearbox.
(reduce speed & multiply torque)

- If ω is at 3500 rpm & wheel rolling radius is 0.3m
+ what is the speed of vehicle?

Torque production

ω delivers speed & Torque

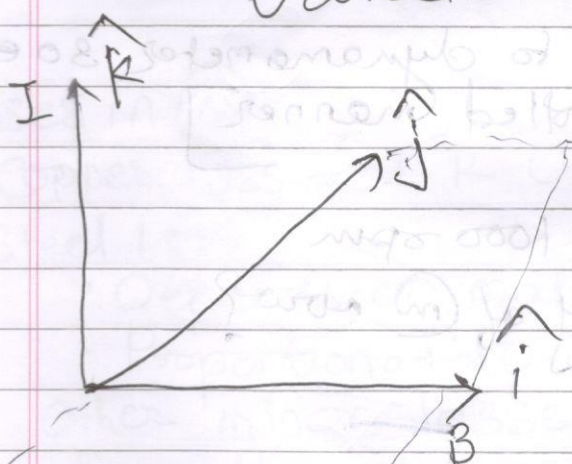
Force on a wire



$$\vec{F} = \vec{I} l \times \vec{B}$$

vector
cross
product

↓ direction of force $\vec{F}$



$$\vec{I} = I \hat{j}$$

$$\vec{B} = B \hat{i}$$

$$\vec{F} = B I l (\hat{j} \times \hat{i})$$

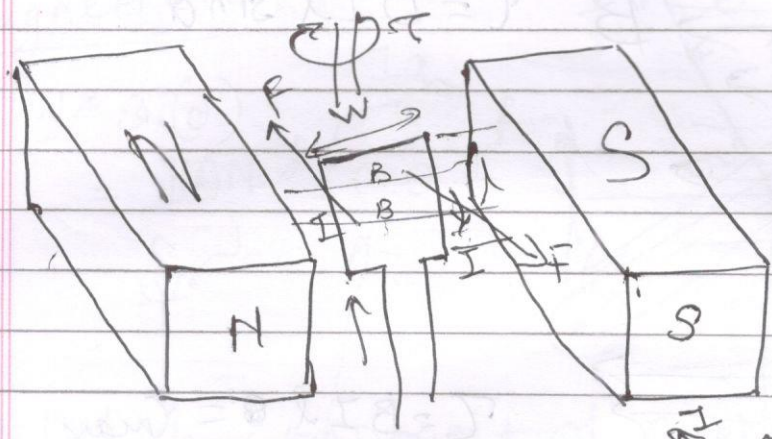
$$= B I l (-\hat{k})$$

$$|\vec{F}| = B I l \text{ magnitude}$$

direction - $-\hat{k}$ dirⁿ

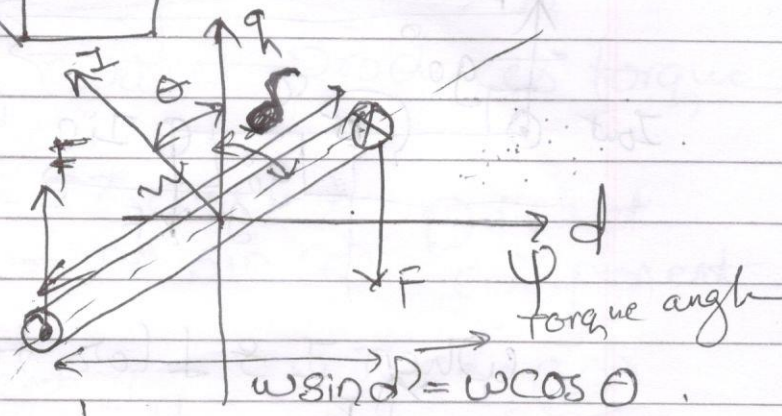
Which is downwards

Torque on a loop.



This is PMDC @ involving split rings & sliding carbon brushes for commutation

Torque $|\vec{\tau}| = F * W * \sin(\delta)$
 δ , the torque angle
 $\delta = 90^\circ - \theta$
 $l * w = A$



This is complement of δ

$|\vec{\tau}| = B I l * w * \sin(\delta)$
 $|\vec{\tau}| = B I A * \sin(\delta)$
 $|\vec{\tau}| = B A N * I \sin(\delta)$
 $N = \text{no. of turns}$

If I look from above
 • → current ~~entering~~ coming out
 X → current leaving entering
 τ in clockwise dirⁿ

$|\vec{\tau}| = \psi * I * \sin(\delta)$

(BAN) Flux linkages

phase advance angle $\theta = 90^\circ - \delta$

$\vec{\tau} = \vec{I} \times \vec{\psi}$

$|\vec{\tau}| = I * \psi * \sin \delta$
 $= I * \psi * \cos(\theta)$

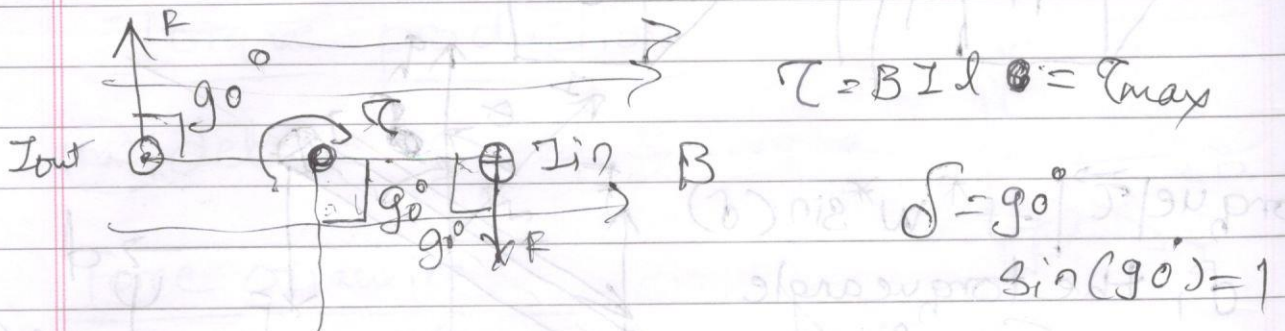
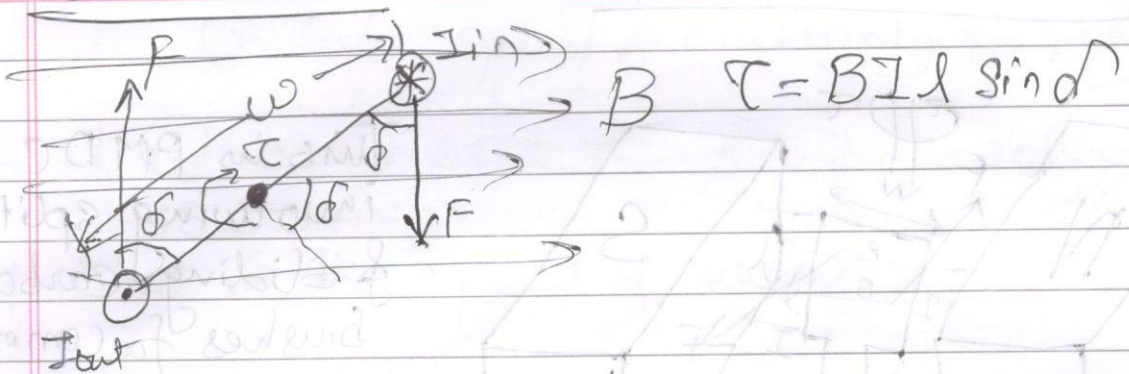
+ How to increase torque

$\uparrow B \rightarrow$ better magnet high price

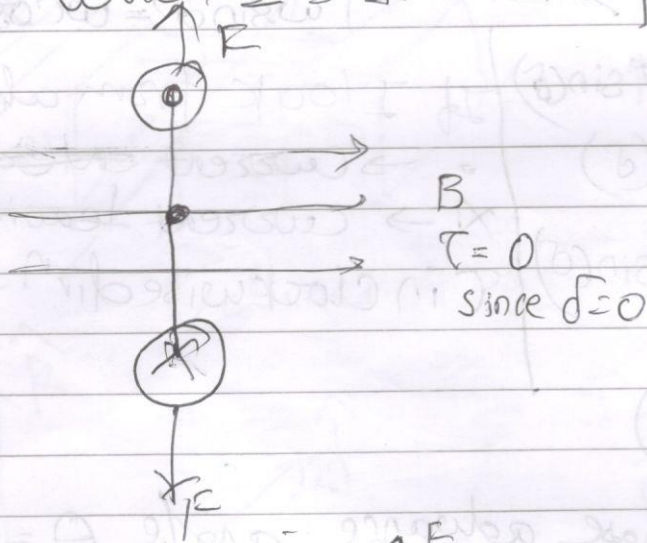
$\uparrow I \rightarrow$ torque by $\uparrow I$

$\uparrow A \rightarrow$ Bigger @ inefficient & costlier

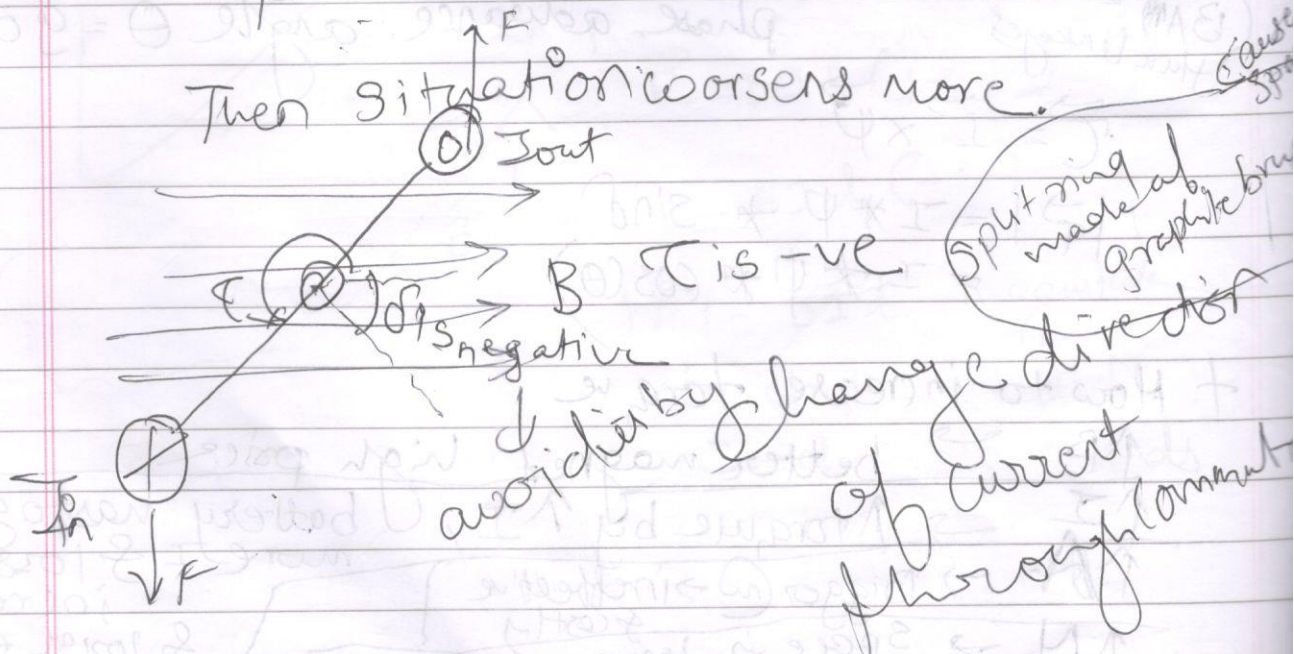
battery has to give more I & losses increase

Commutation.

When $I \perp B$ or $\delta = 90^\circ \rightarrow \tau_{\max}$.



Then situation worsens more.



Torque T_{mag} created by magnet

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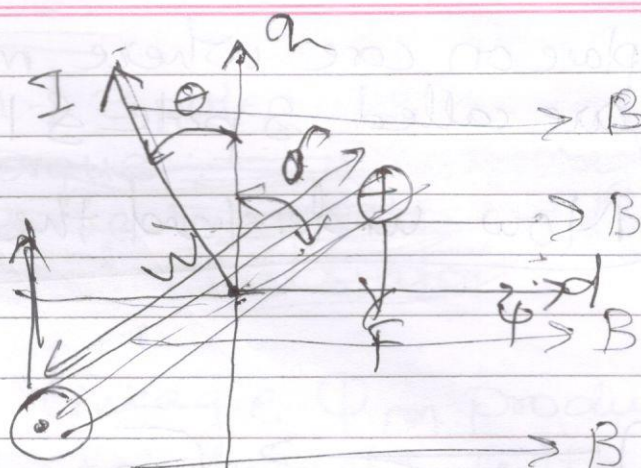
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magnetic torque $\rightarrow$

$$F \cdot W \cdot \sin(\phi) = BAN \cdot I \cos \theta$$

ϕ



+ Which component of current produces torque

1st component $I_d = I \cdot \cos(\phi)$ Direct component
 $= I \cdot \sin(\theta)$



2nd component $I_q = I \cdot \sin(\theta)$

Quadrature component
 (magnetic torque component)

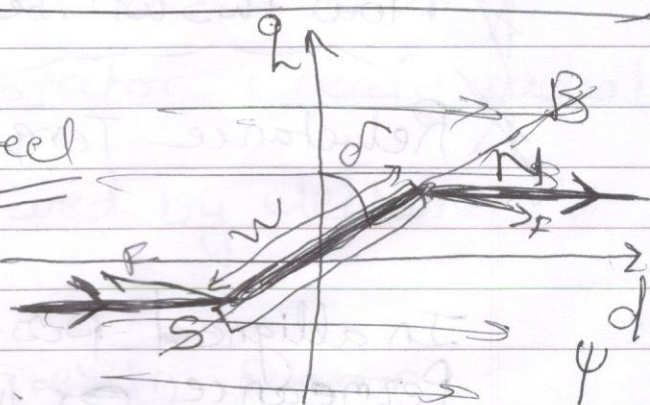
Causes torque

~~If I ignore Curvature & keep only steel core~~

Winding is wound on steel not in air

Imagine it was just an iron loop no current flowing

* will it rotate & why.

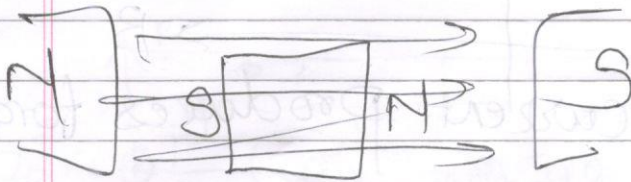


$\rightarrow$ Mag. flux flows more through iron due to its lower reluctance. This gets induced in iron & acts as a magnet

more magnetic lines flow through steel than air because it gives less reluctance path

place on core where magnetic lines enter are called S pole & leaving are called N pole

Now understand the construction



So this south pole ~~will~~ will get attracted toward N pole of stator & North pole will get attracted towards south pole of stator magnet which will cause upward force F

Now this will be reluctance torque.

$$\begin{aligned} \text{Reluctance Torque} &= R^* W^* \cos(\delta) \\ &= F^* W^* \sin(\theta) \end{aligned}$$

In aligned position, reluctance is min.
Permeance or inductance is max.

magnet free Stepper motor
which works on principle
of reluctance torque

FV @S & controllers EV @S & controllers

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PMSM: — IPM - 3 pole pairs.

~~Magnetic torque: — When magnetic rotor rotates the magnetic field is at & flux is linked with stator~~

magnetic flux linkage Ψ_m produces a torque when it interacts with the $\perp$ lar component (q -axis) of winding current.

$$\begin{aligned}\text{Magnetic torque, } \vec{T}_m &= \vec{I} \times \vec{\Psi}_m \\ &= -I_q * \Psi_m \\ &\equiv \text{Same as in PMDC}\end{aligned}$$

Windings are in the stator

- Rotating wires in rotor are messy — best avoided.
- Instead, loops in stator create radial magnetic fields.
- Commutation is achieved by electronic swing.

Permanent magnets are in Rotor

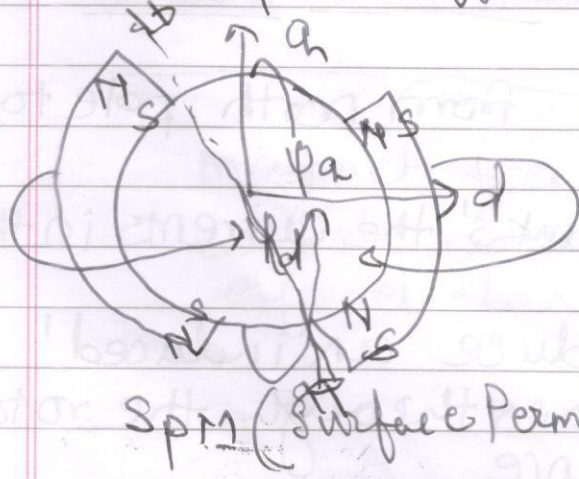
- Magnetic flux flows from north pole to successive south pole.
- The magnetic flux 'links' the currents in the winding.
- The windings produce an 'induced' flux too — that flows through the rotor that offers reluctance.

- windings are wound around tooth
- windings has current in it.
- that steel tooth becomes magnet.
- flux from this tooth will flow to some other tooth with opposite polarity
- Magnetic field produced by magnet interacts with ^{current in} windings & it produces magnetism (we talked about it)
- But this magnetic field due to ~~rotor~~ ^{stator} tooth ~~has~~ ^{has} no winding to interact with & it passes through rotor where it experiences reluctance.

[So there is additional magnetic flux apart from magnetic flux of permanent magnet]

Reluctance : —

- The winding current also produces flux
- The flux flows through the principle dirⁿ in rotor - direct & quadrature.
- Each path offers reluctance to flow of flux.



In an SPM (N)

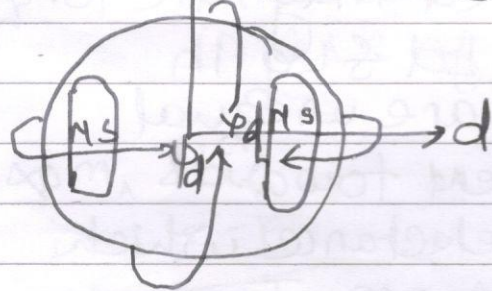
- Reluctance path is either through the magnet
- But the permeability of magnet and air are almost equal.
- Hence both paths offers same reluctance.

Now in case of spm consider two tooth of stator slots on either side of rotor.

- The flux has to go through steel.
- Now if for computational purpose we consider 2 components of this flux
- one across d axis
- one across q axis
- Now the flux travelling along q axis has to encounter air & steel (diameter)
- where as for d axis it has to encounter air, magnet & steel (reluctance)
- air & magnet has almost same reluctance & it is very high.
- So reluctance is higher along d axis almost.

L_d & L_q are same

For IPM though



Opposite to SPM.

Reluctance along d axis is very high than that in q axis.

IPM (Internal permanent magnet)

★ reluctance = $\frac{1}{\text{permeance}}$

& $\frac{1}{\text{permeance}} \times N^2$ is called Inductance

Permeance $\times N^2 =$ Inductance

L_q is high
 L_d is lower

The induced winding flux linkage $\Psi = L \cdot I$

Thus,

— along magnet, $\Psi_d = L_d \cdot I_d$. d axis

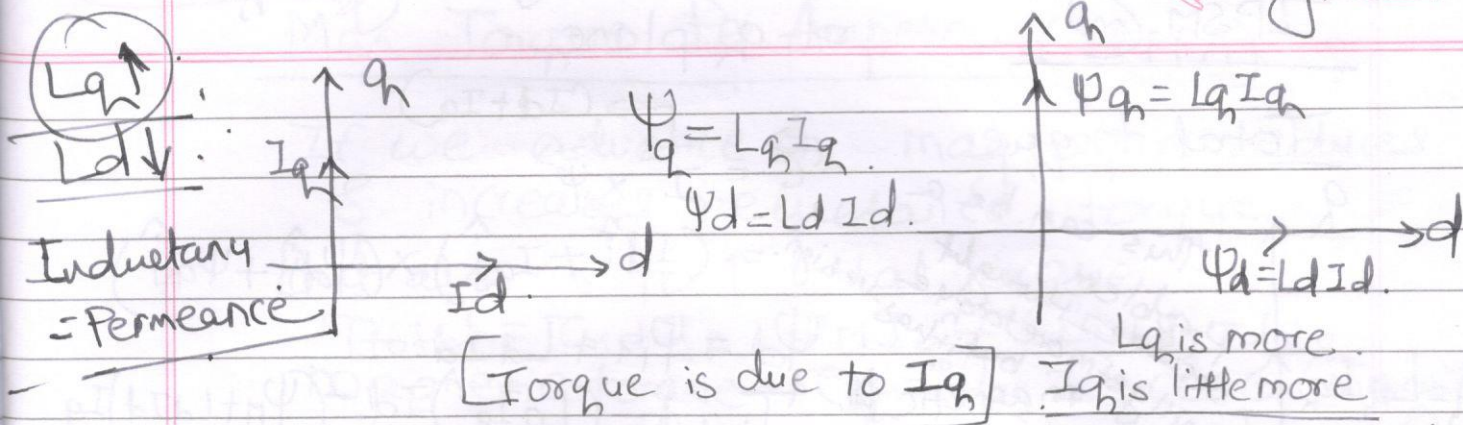
— Perpendicular to magnet, $\Psi_q = L_q \cdot I_q$. q axis

Consequence of this reluctance

- Reluctance in one path is different from reluctance in other path.
- $\&$ Steel will try to align itself to path of least reluctance.
- In other words it will try to align itself to max. inductance.
- This will cause a torque.

So far we have learned magnetic torque but now we have L_d & L_q in different direction & are unequal will result in alignment towards max inductance or least reluctance which will be called reluctance Torque. It is like Baker's Dozen.

In cross product, 2 angles or dirⁿs are opposite means if $\underline{\Psi_d} \times \underline{I_q}$ or $\underline{\Psi_q} \times \underline{I_d}$ they will have opposite angle dirⁿ.



$$\vec{\tau} = \vec{I} \times \vec{\Psi} \quad \{\text{vector cross product}\}$$

$$\Rightarrow \vec{\tau} = (I_d \hat{i} + I_q \hat{j}) \times (\Psi_d \hat{i} + \Psi_q \hat{j})$$

magnetising torque
is only given by I_q

but Reluctance torque
is given by both I_d & I_q .

$$= (I_d \Psi_q - I_q \Psi_d) \hat{k}$$

Both components of current contribute to Reluctance Torque.

$$\text{Thus } \tau_r = |\vec{\tau}| = (L_q - L_d) I_d I_q$$

Note negative sign! $L_d = L_q$ in SPM less torque

$L_d \neq L_q$ in IPM increase torque

$L_q / L_d = \text{Saliency of motor.}$

in case of SPM it is 1 (No reluctance torque)

in case of IPM it is more ~~than~~ than 1
(A lot of reluctance torque)

$$\hat{i} \times \hat{j} = -\hat{k}$$

So in IPSPM (M) we deliberately make I_d -ve.
 So reluctance torque adds to magnetising torque.

IPSPM (M) $d-q$ plane.

Total torque $\vec{T} = \vec{I} \times \vec{\Psi}$

This can be a problem because I thought I will get more torque but it has -ve sign.

$$\vec{T} = (\hat{I}_d + \hat{I}_q) \times (\hat{\Psi}_d + \hat{\Psi}_q)$$

$$\Psi_d = \Psi_m + L_d I_d$$

$$\Psi_q = L_q I_q$$

$$T_{total} = (L_q I_q) I_d - (\Psi_m + L_d I_d) I_q$$

$$T_{total} = T_m + T_R$$

$\Psi_m + L_d I_d$ $\rightarrow$ d component of reluctance flux.
 $\hookrightarrow$ permanent magnet flux

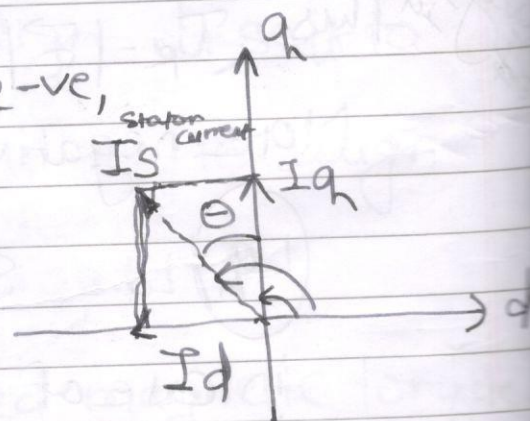
I want to change sign of 2nd term without changing sign of 1st term.
 Since $L_q > L_d$, $(L_q - L_d) \rightarrow +ve$
 T_R adds to T_m only if I_d is -ve.

Phase Advance Angle

Now that we know, I_d must be -ve, for reluctance torque to add to magnetic torque, rather than act against it.

$$I_s^2 = I_q^2 + I_d^2$$

$$\tan \theta = \frac{I_d}{I_q}$$



Current has to be 180° to magnetic field but

Since we have advanced the θ ,

Previously we know, at $\theta = 90^\circ$ torque is max now we are greedy for reluctance torque we make $\theta > 90^\circ$.

$$I_q = I_s \cos \theta$$

θ is called phase advance

Max Torque Per Ampere (MTPA)

If we advance θ , mag torque reduces & increases reluctance torque.

But aim is to $\uparrow$ total torque.

$$T_{\text{total}} = T_M + T_R = \Psi M I_a + (L_q - L_d) I_d I_q$$

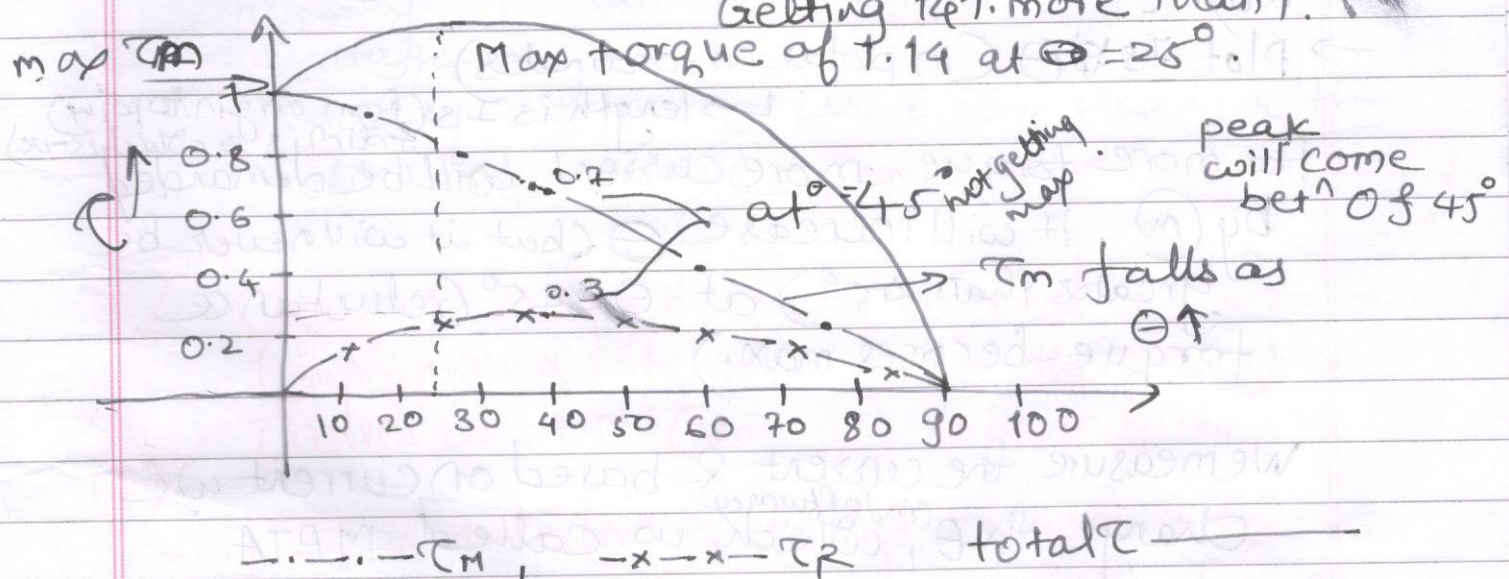
magnetic torque. $T_M = \Psi M I_s \cos \theta$ (max when $\theta = 0$)

$$T_R = (L_q - L_d) I_s^2 \cos \theta \sin \theta.$$

$$= \frac{1}{2} (L_q - L_d) I_s^2 \sin(2\theta) \quad (\text{when } \theta = 45^\circ)$$

$$T_R \propto I_s^2$$

Getting 14% more than 1.



For optimum value of θ .

- The sum of 2, which is the total torque, will be max. at some angle betⁿ 0 & 45° .
- That is the angle at which max torque is obtained, for a given value of I_s .

Reluctant torque $\propto I_s^2$ peak value $\propto I_s$
magnetic torque $\propto I$ of torque

If I half the current, T_R will fall drastically & T_M will also reduce.

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At higher current Δe will be larger
at smaller current Δe will be smaller

$$T = A * I_s * \cos(\theta) - B * I_s^2 * \sin(2\theta)$$

ψ difference in inductance ($L_d - L_q$)

* Assume $A = 0.01 \text{ m}^2$
 → Vary I_s from 0 to 100A (in increments of 10A)
 For each value of I_s , find the Θ that gives
 max τ .

length is $\sqrt{x^2 + y^2}$ (from origin to point (x, y)) which is

We measure the current I based on current we change θ ^{in software}, which is called MPPT.

If I select ϕ correctly, then I get more ~~current~~ torque than at any other angle for same current.

Speed and back emf.

Torque related current.

Speed? we talked about current but not Voltage

Faraday :- whenever flux changes, it produces ~~changes~~ emf (fancy word for voltage)

Greater the rate of change of flux, greater will be magnitude of emf. & this emf will oppose its cause that is change in flux [dir^n]

EMF will suppress change in flux.

Faraday 1st law deals with magnitude of Voltage

Faraday 2nd law or Lenz's law deals with dir^n of voltage [conservation of energy]

Change in magnetic field around a wire (by rotating the rotor inside the stator) causes an EMF (voltage) to be induced.

[If I take m & do not connect it to power supply & instead connect wires to oscilloscope, then & then rotate the rotor then on oscilloscope I see, sinusoidal voltage waveform] $\rightarrow$ back emf profile

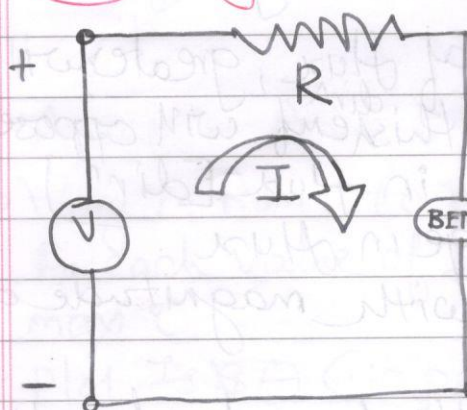
If I rotate rotor faster then amplitude (y-axis) & freqⁿ (x-axis) will also increase

$\rightarrow$ The dir^n of this EMF would be such that it would oppose the change

- An opposing voltage, if rotation is causing an applied voltage.

- The higher the speed, greater is the rate of change in magnetic field, & higher the back emf

(M) Dynamics (simple PM-DC (m))



⇒ when voltage is applied to (m)

- 1) Current flows, producing torque
- 2) The (m) speeds up
- 3) This speed creates B-EMF
- 4) The B-EMF causes current to reduce

$$V - \text{BEMF} = I \cdot R$$

diff ↑ more I

diff ↓ less I

5) This reduces torque

→ (m) at rest - no flux produced - no back emf
 So large current will flow
 large current ↑ torque, speed ↑, BEMF ↑
 as BEMF ↑, difference betⁿ BEMF & voltage which reduces I, which will reduce torque

(so initially (m) started as high torque (m) now is running with lesser torque)
 & Then steady state will reach.
 (speed is neither ↑, nor ↓)

In the beginning I applied 5V, now (m) has 4V & running at 1000 rpm
 $5V - 4V = 1V$ will cause the current, which produces torque which will overcome the

So Speed (Steady state speed) at which load torque is supplied by (m) torque is called Steady state speed.

→ It depends on voltage applied & Load connected.

→

* A vehicle is cruising at 36 kmph, applied voltage is 25V, $BEMF = 24V$ & winding resis is $20m\Omega$ (PMDC (m))

→ How much Current (m) is drawing

→ What is the power delivered to vehicle.

→ What is load torque due to vehicle, reflected on (m)

→ What is efficiency of (m)

→ If the applied voltage were increased by 1V, What will be final speed of (m)

(neglect losses).

* Speed [electrical & mechanical]

BEMF wave form is completed

← When I cover (mechanically 120° , magnetically 360°)
Electrical cycles are P times mechanical cycles
No. of P poles.

mechanical speed to electrical speed. (Pole pairs P)

Consider a rotor with P pole pairs.

→ when it rotates once, the BEMF waveform would have completed P cycles.

→ There are P numbers of $d-q$ axis pairs.

$P=3$
angle betn
d & q is 30°
got $\alpha=30^\circ$

when I complete 1 mechanical rotation, I am ~~com~~ cutting 3 pole pairs

→ Hence in 1 mechanical rotation is equivalent to P electrical rotations

→ If the mechanical rotation speed is ω_m , the electrical speed is $\omega_e = P * \omega_m$.

$$\rightarrow \text{Electrical frequency } f_e = \frac{\omega_e}{2\pi}$$

$$= \frac{P \omega_m}{2\pi}$$

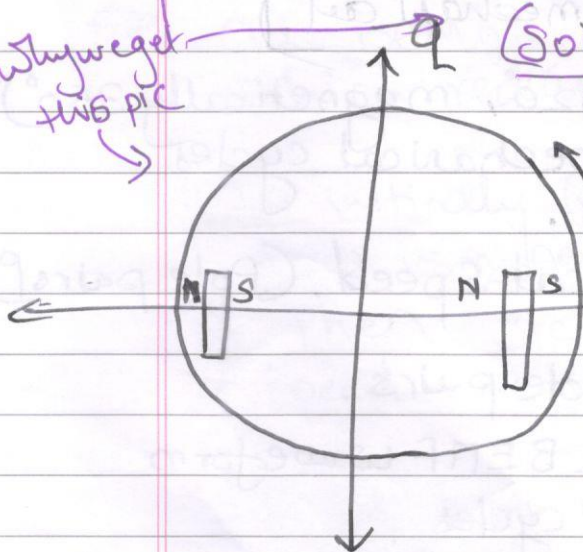
$$f_e = P * \text{rps}$$

consider fig. showing previously

The BEMF Constant: K_e

[Grabbed ends of d axis, it was forming angle of 60° ; Grab it & stretch it out 3 fold so everything unfolds & success pole pairs overlaid on itself p-times, as belows ^{under each other}

Why we get this pic



(So it is like unwrapping & making it go round p times)

faster the rotor rotates - more BEMF

$$\omega_e = P * \omega_m$$

Now relation betⁿ BEMF & Mech speed.

$$\text{BEMF} = (\text{some const}) \times \text{mech speed}$$

$$\downarrow$$

$$\frac{\text{BEMF Voltage}}{\text{Mech Speed}}$$

Speed $\rightarrow$ freqⁿ of BEMF voltage

magnitude $\rightarrow$ how high waveform comes from BEMF constant.

We know that BEMF is proportional to ω_m , the mechanical speed

$$\boxed{\text{BEMF-const, } K_e = \text{BEMF}/\omega_m}$$

The BEMF-const can be measured in
volts $\leftarrow V/(\text{rad/s}) = V\cancel{s}/\text{rad} = V\cancel{s}/\text{rad}$.

$V/(1000 \text{ rpm})$ {common industry practice}

In both cases, the mechanical speed is considered as the basis for measurement.

A 8 pole m is run at 3000 rpm and the measured back-emf is 18V.

→ what is the frequency of back emf voltage?
→ what is K_e of m in

a) $V/1000 \text{ rpm}$

b) $V\cancel{s}/\text{rad}$?

1-60s

$$\frac{3000 \text{ rpm}}{60} =$$

$$\frac{18}{2} \times 8 = F_c = \frac{P \omega_m}{2\pi}$$

$$= \frac{8 \times 3000}{2\pi}$$

$$= \frac{12000}{3.14}$$

$$= 40$$

$$K_e =$$